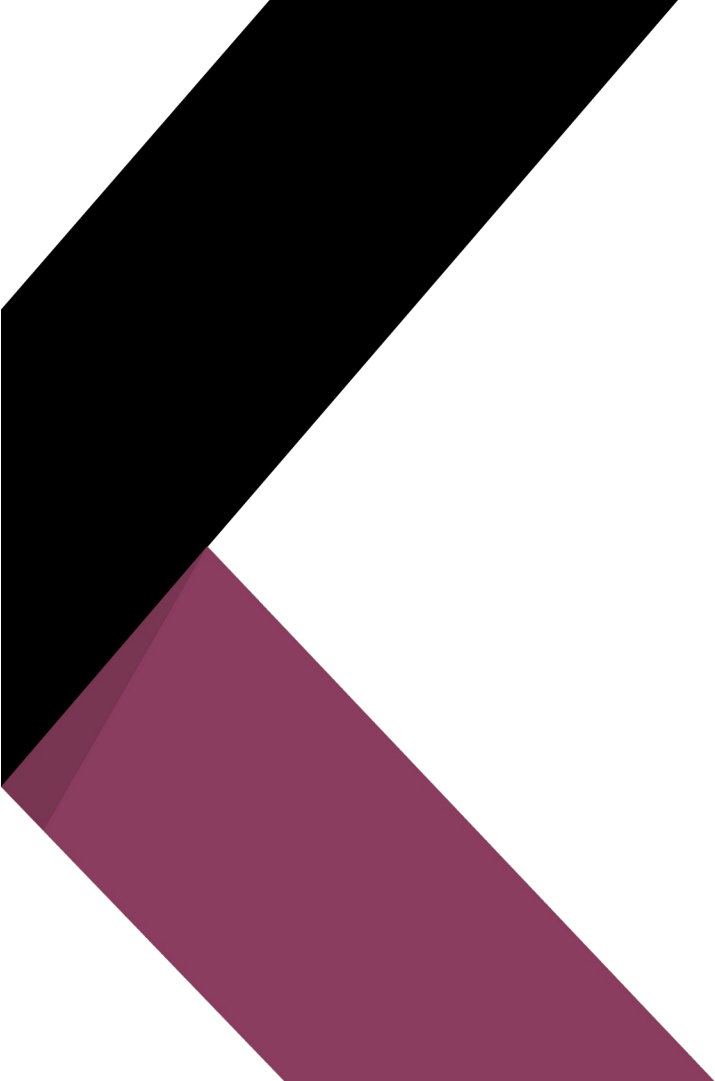


Electric Machines

A decorative graphic on the right side of the slide. It consists of a large black triangle pointing downwards, which is partially overlapped by a smaller maroon triangle pointing upwards. The maroon triangle is positioned such that its top vertex is inside the black triangle, creating a layered effect.



Introduction

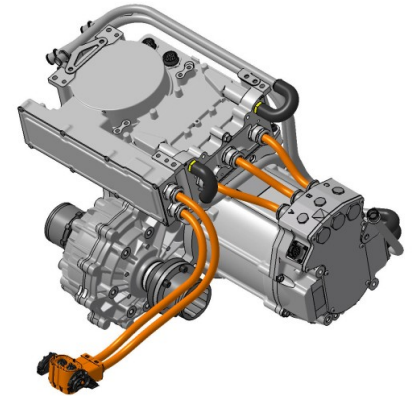
Electric Machines - Overview

- Electric machines play a crucial role in all industry fields and power generation, as well as in automotive traction. In this modern reality and under the global need for CO₂ emission reduction, electric motors and generators are continuously evolving, aiming for higher efficiencies simultaneously with more reliable operation.
- Motor drives are the core technology for EVs that convert the on-board electrical energy to the desired mechanical motion. Meanwhile, electric machines are the key element of motor drive technology. The requirements of electric machines for EVs are much more demanding than that for industrial applications.



The general requirements of electrical machines for automotive are much more stringent than those for industrial applications:

- High torque density and high power density;
- Wide speed range, covering low-speed creeping and high-speed cruising;
- High efficiency over wide torque and speed ranges;
- Wide constant power operating capability;
- Low acoustic noise;
- Reasonable cost.

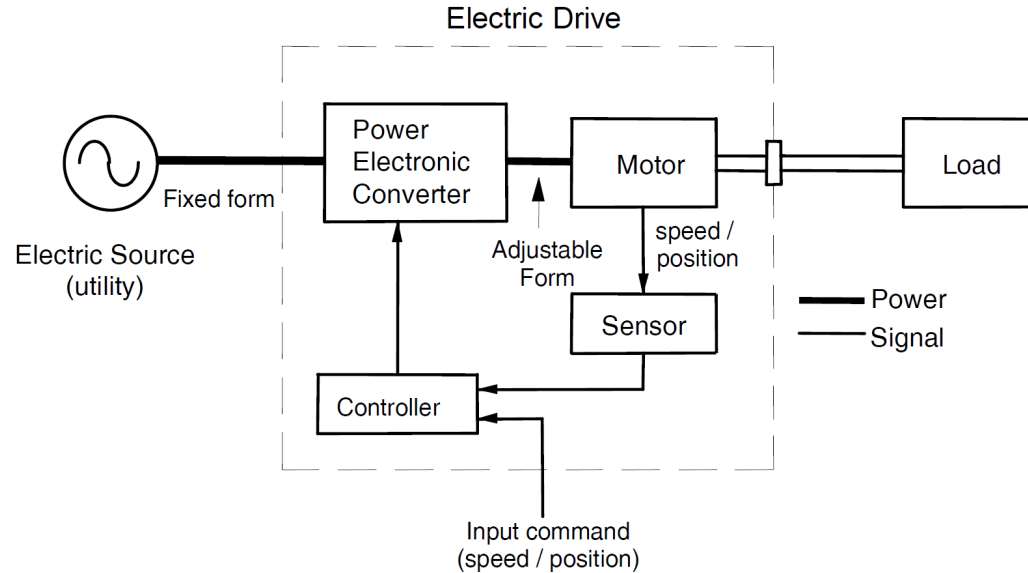




Electrical machines

Electric Drive

An electric drive can be defined as an electromechanical device for converting **Electrical Energy into Mechanical Energy** to impart motion to different machines and mechanisms for various kinds of process control.



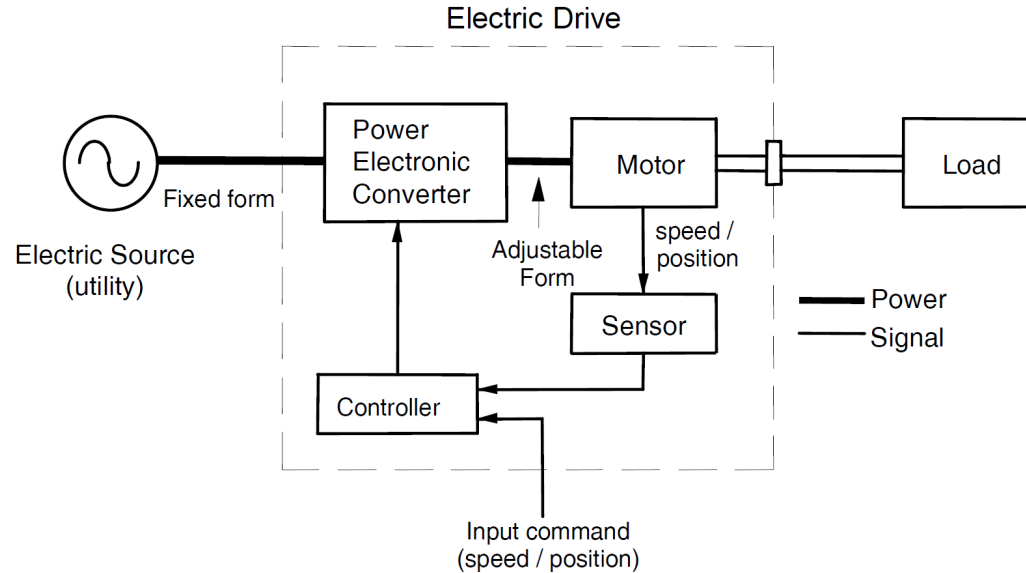


Electrical machines

Electric Drive

The basic architecture of a motion control system includes:

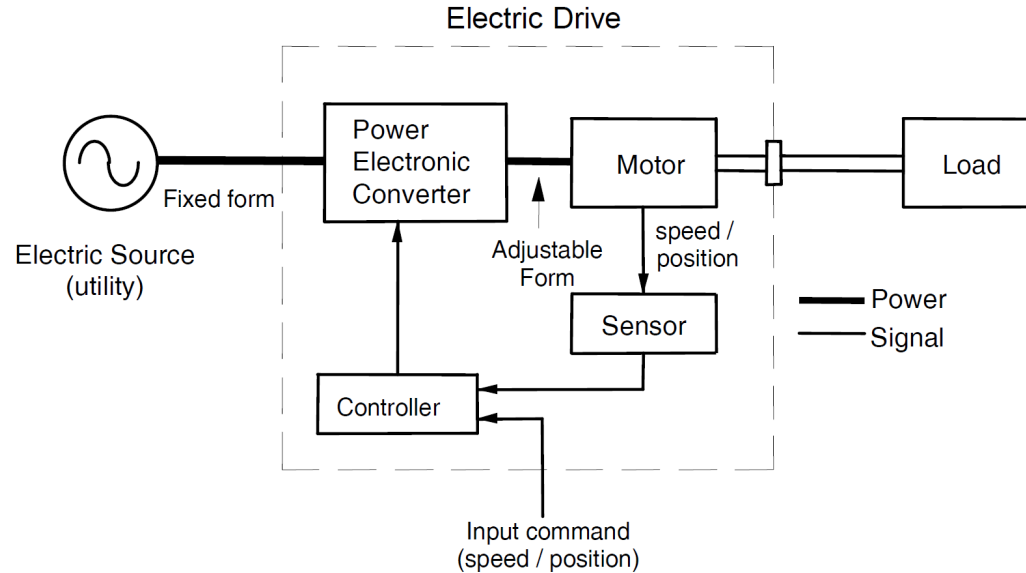
1. **Electric Source:** provides the energy to the drive system
2. **Power Electronic Converter:** interfaces the motor with the power source and provides the motor with adjustable voltage, current and/or frequency.
3. **Controller:** by comparing the input command for speed and/or position with the actual values measured through sensors, provides appropriate control signals to the power converter.





Electrical machines **Electric Drive**

4. Electric **Motor**: selected according to the power level, environmental factors and performance required by the load.
5. One or more feedback **Sensors** to return the position and/or velocity of the actuator to the motion controller in order to close the position and/or velocity control loops.
6. **Mechanical components** to transform the motion of the actuator into the desired load motion, including: gears, shafting, ball screw, belts, linkages, and linear and rotational bearings.



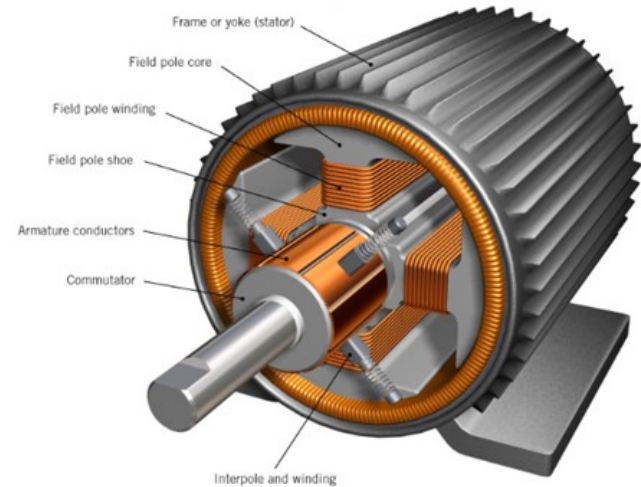


Electrical machines **Electric Motors**

An electric motor converts electrical energy to mechanical energy. It consists in a cylindrical structure with a fixed, static part, called **stator**, and a moving, rotating one, the **rotor**.

Both the structures can be populated with either **windings** or **permanent magnets**, achieving different electrical and magnetic performances.

The energy conversion happens in the **interaction between the magnetic fields** of the two structures, which produces a net output torque.





Basic principles



Magnetic flux, Φ , can be interpreted as a number of magnetic lines of force that cross certain given area - the greater the number of lines, the greater the flux and stronger the field.

Magnetic force

A current carrying conductor in presence of a magnetic field is exposed to a force equal to:

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

Induced electromotive force

Consider a rectangular loop rotating in a uniform magnetic field B .

Time-varying flux causes an induced electromotive force (e.m.f.), e , at terminals of a closed loop (or, a single conductor):

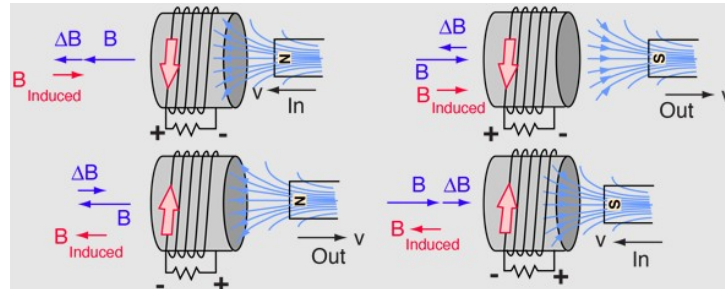
$$e = -\frac{d\Phi}{dt}$$



Basic principles Electromotive force

In a motor, the induced emf always acts to reduce the current in the loop. That's why it is called “**back emf.**”

In fact, Faraday-Neumann-Lenz law establish that *the induced current in a loop is in the direction that creates a magnetic field that **opposes the change** in magnetic flux through the area enclosed by the loop.* The induced current tends to keep the original magnetic flux through the circuit from changing.





Basic principles

Torque

Torque

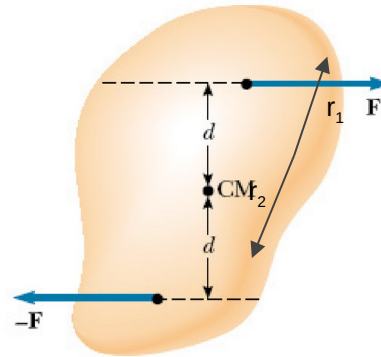
A ***couple*** is a pair of forces, $\mathbf{F}$ and $-\mathbf{F}$, equal in magnitude, opposite in direction, and separated by a distance, d . It can cause a rotation.

The moment of a force $\mathbf{F}$ about a point O is the vector product of $\mathbf{F}$ and the position vector, $\mathbf{r}$, from O to the point of action of the force vector $\mathbf{M} = \mathbf{r} \times \mathbf{F}$

The moment of a couple:

$$\mathbf{M} = \mathbf{r}_1 \times \mathbf{F} + \mathbf{r}_2 \times (-\mathbf{F}) = (\mathbf{r}_1 - \mathbf{r}_2) \times \mathbf{F} = \boldsymbol{\tau}$$

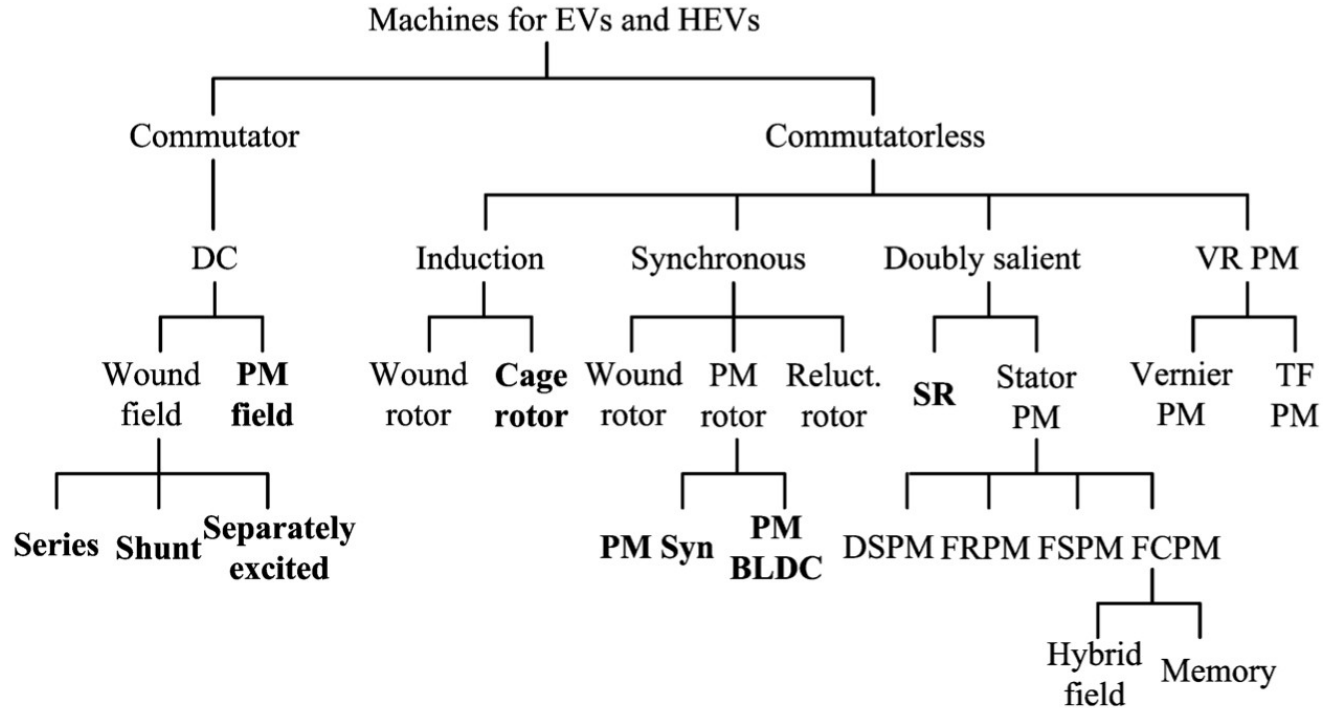
is called ***torque***, $\boldsymbol{\tau}$.





Electrical machines

Classification





DC Machine



DC Machine **Pros and Cons**

Pros:

- Simple operation and motor control
- Cheapest and simplest motor
- Speed proportional to voltage

Cons:

- Requires high maintenance
- Sparking
- Low-life span (due to physical wear on brushes)



DC motor drive technology is very mature and well developed. However, with the ever-increasing demand on high efficiency, high power density, and maintenance-free operation, DC motor drives are **no longer attractive** for EV propulsion.

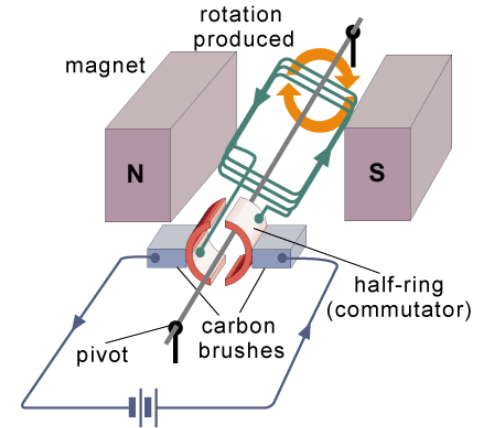


DC Machine Structure

The **stator**, external, equipped with salient poles, works as an inductor and can be realized:

- with **excitation windings** arranged around the pole bodies; such windings are crossed by direct current and are connected to each other in controversial series to generate magnetic polarity of alternating sign from one pole to the next;
- with **permanent magnets**.

The internal **rotor**, consisting of a cylinder of magnetic material, works as armature; on the periphery of the rotor are uniformly distributed slots where the active conductors are housed. The set of active conductors (connected in series with each other) constitutes a closed circuit.





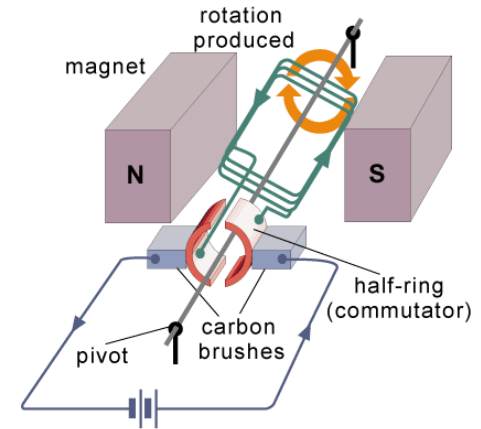
DC Machine Structure

The part in the air between stator and rotor is called **air gap**, in this area the field magnetic has the greatest intensity.

The **commutator** is mounted on the same shaft as the rotor. This is for the purpose of supply the windings with the power supply voltage and ensure that the torque generated by the current that passes through is constant or at least unidirectional.

The **brushes** crawl on the collector and electrically connect it to the external terminals of the machine.

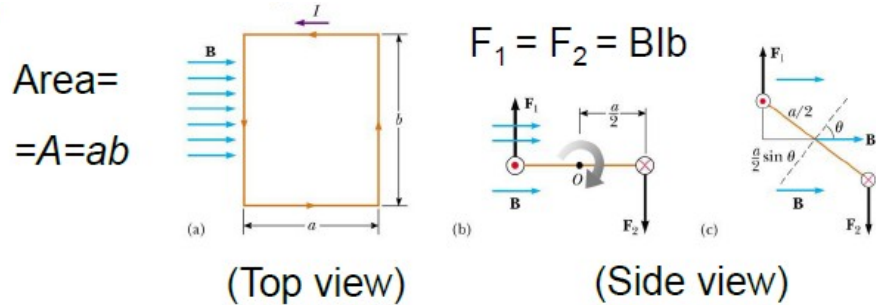
The presence of the system commutator-brushes causes various issues, mainly the continuous maintenance required and the undesired sparking.





DC Machine Working principle

Consider a rectangular conducting loop carrying current I in the presence of a magnetic field:



There is a magnetic force on sides of length b (in opposite directions).

If loop can pivot about its center, these forces provide a (clockwise) **torque** about the pivot point.

Maximum torque produced is $\tau_{max} = B I b \left(\frac{a}{2} + \frac{a}{2} \right) = B I A$

If field makes an angle ϑ with a line perpendicular to the plane of the loop then $\tau = B I A \sin \vartheta$

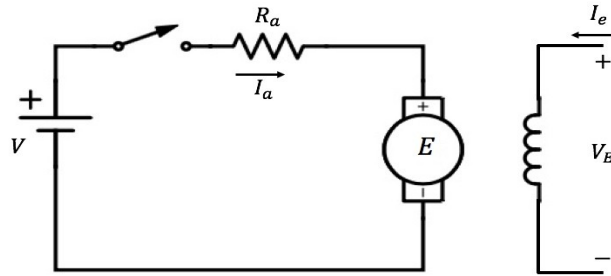
For any loop shape having N turns: $\tau = N B I A \sin \vartheta$



DC Generator



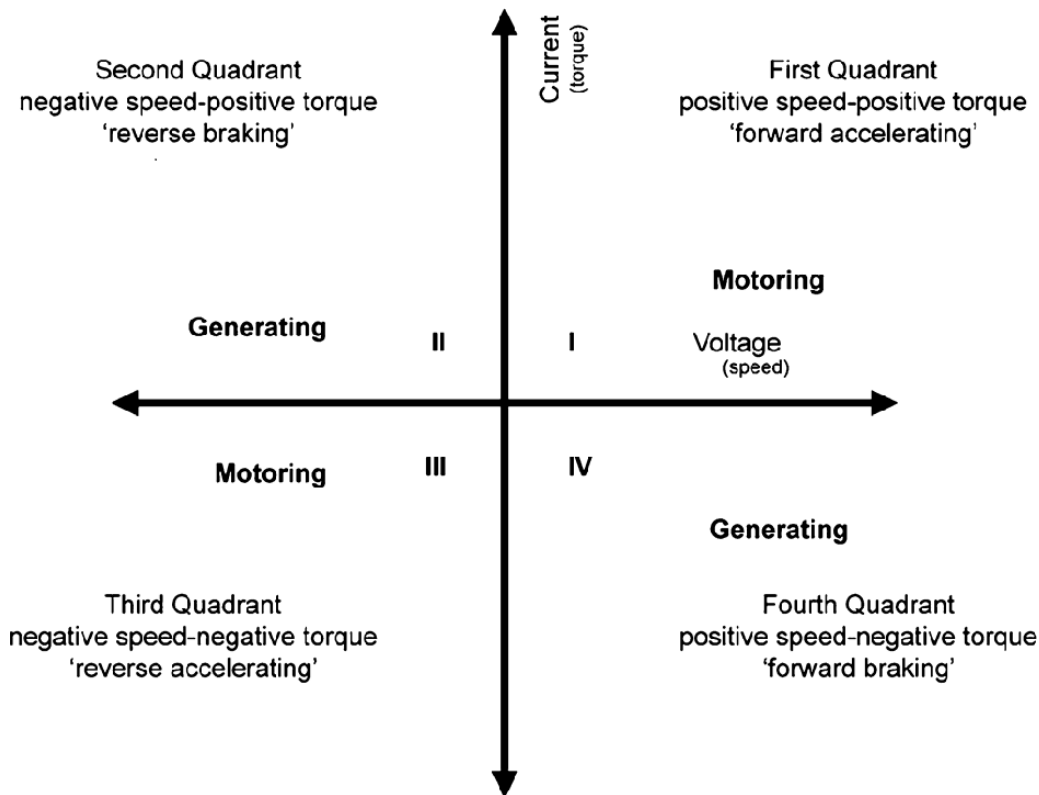
DC Machine Braking



If the directions of the voltages and currents are those shown in the figure, the machine acts as a motor (switch closed). If the E is made greater than V , the armature current is reversed, the excitation flux has remained with its polarity and the machine switches to operate as a **brake** because the torque reverses becoming negative and instead of being a driving force it becomes braking. Therefore, the machine acts as a brake towards the load and from the electrical point of view it acts as a **generator**, takes energy from the load, converts it into electricity and this can be dissipated or recovered.



DC Machine 4 Quadrants





Inverter

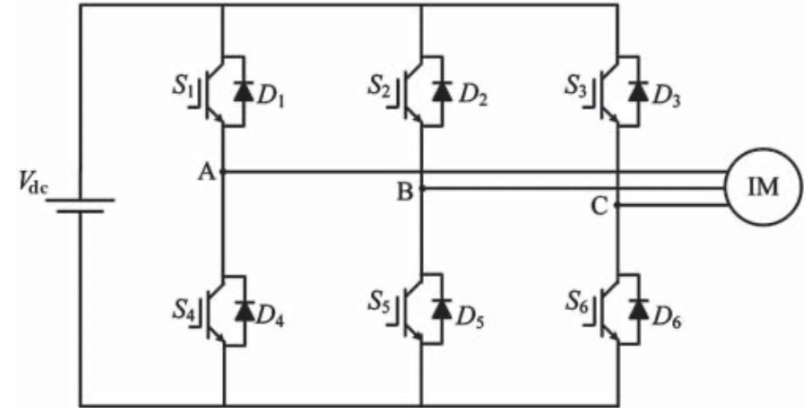


Inverter Basic principle

The inverter design highly depends on the technology of power devices. Nowadays, the **insulated-gate bipolar transistor** (IGBT)-based inverter is most attractive and has been widely accepted by modern EVs.

The selection of power devices is based on:

- **the voltage rating** is at least twice the nominal battery voltage;
- **the current rating** is large enough so that there is no need to connect several power devices in parallel;
- **the switching speed** is sufficiently high to suppress motor harmonics and acoustic noise.





Inverter Basic principle

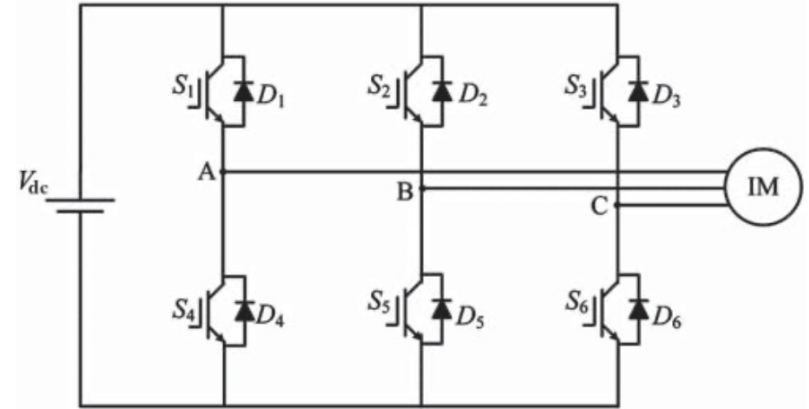
Inverters for Induction or Permanent Magnet motors are generally classified into **voltage-fed** and **current-fed** types.

Because of the need of a large series inductance to emulate a current source, the current-fed inverters are rarely used for EV propulsion.

In fact, the voltage-fed inverters are almost exclusively used because they are very **simple** and can have **power flow in either direction**.

Control strategies can be classified as:

- **PWM** (Pulse Width Modulation);
- **Six-step**.





Inverter PWM Control

The structure of the PWM inverter coincides with that of the 6-step inverter, what changes is the switching control technique of the various static switches.

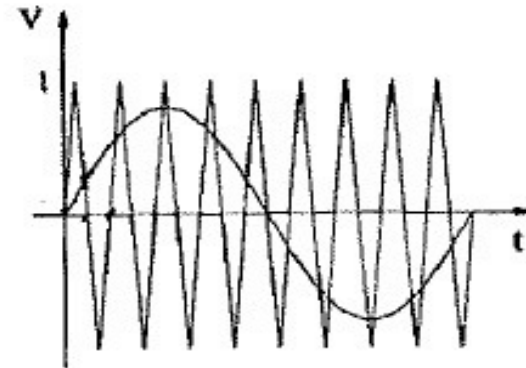
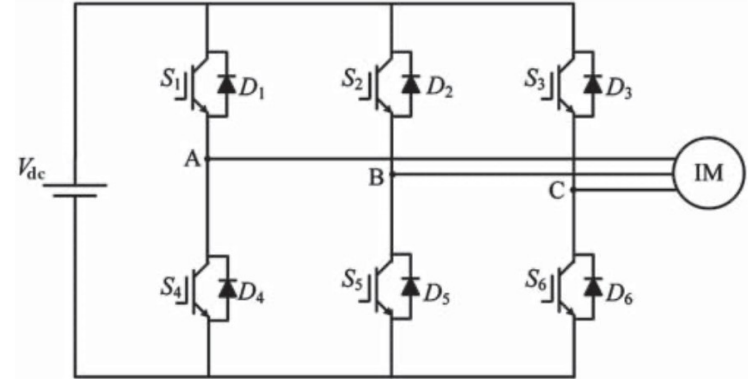
Comparison between:

- **Modulating sinusoid** of amplitude A_s and frequency f_s
- **Triangular carrier wave** of constant amplitude A_t and frequency f_t with:

$$A_t > A_s$$

$$f_t \gg f_s$$

Therefore, you do not equal pulses, but **modulated in width** (PWM).

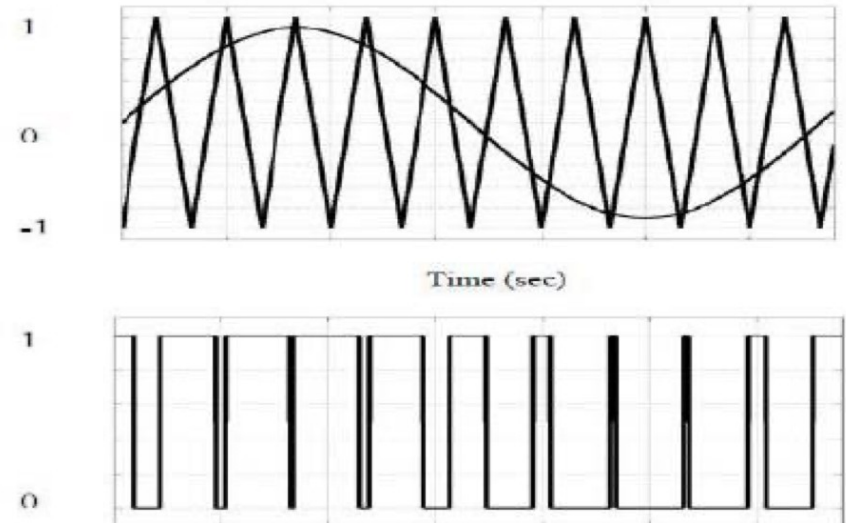




Inverter PWM Control

When the value assumed by one of the modulators exceeds that of the triangular carrier, the switch S of the corresponding phase is turned on while when the value assumed by one of the modulators is less than that of the triangular carrier it is turned off (and the other switch of the same branch is turned on).

The technique can be implemented either in an **analog** or **digital** way.

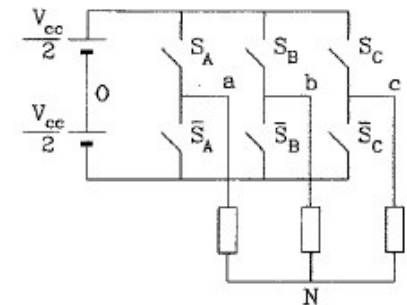
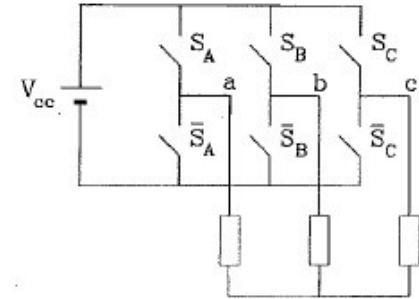
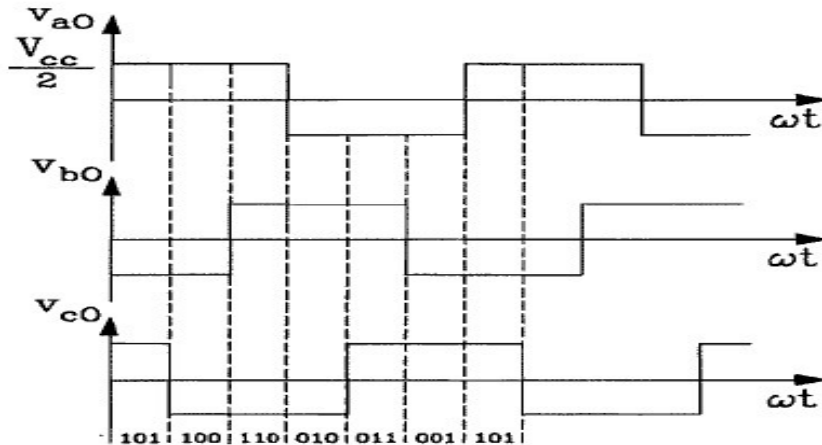




Inverter Six-step Control

In 6-step (or square wave) operation, the switch is alternately closed and open for a time equal to $T/2$, where T is the period of the alternating output voltage.

The switches of the same branch are controlled in a complementary way.





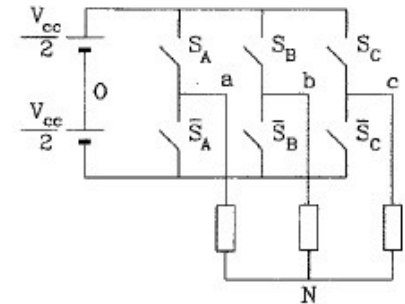
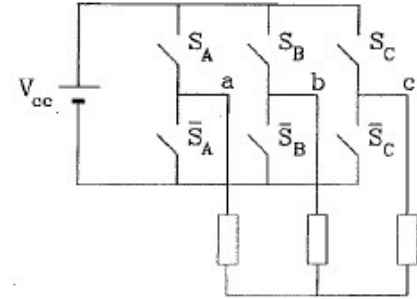
Inverter Six-step Control

In 6-step operation, each switch changes state every $T/2$, while the inverter changes state every $T/6$.

It is called a **6-step inverter** since the waveform is built in the 6 sixths of the period. Every sixth of a period, the components that are commanded for switching change.

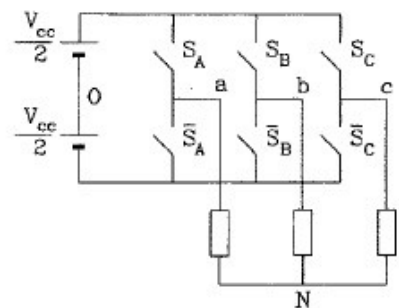
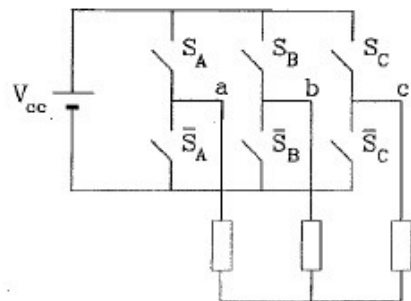
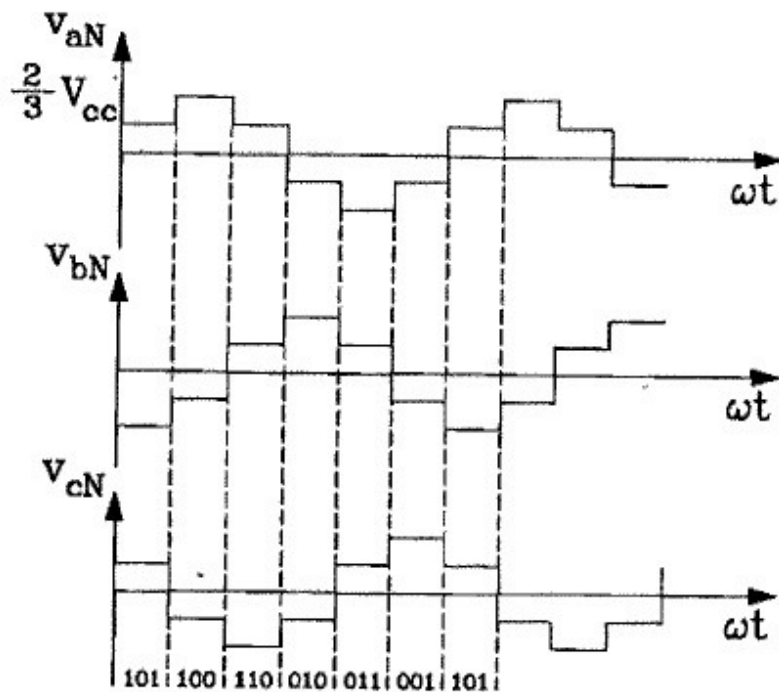
Furthermore, in period T the inverter assumes 6 different states: (101), (100), (110), (010), (011), (001). These are called **active states**.

The amplitude of the output voltage depends only on V_{cc} and the state of the switches.





Inverter Six-step Control





Inverter Requirements

EV propulsion requires the motor drives offering **four-quadrant characteristics** for:

- forward motoring;
- forward regeneration;
- backward motoring;
- backward regeneration.

Particularly, the forward regeneration is essential to **capture the breaking energy** which in turn charges the EV battery, leading to increase the driving range per charge by over 10%.

	Quadrant	Torque	Speed
Forward motoring	I	Positive	Positive
Backward regeneration	II	Positive	Negative
Backward motoring	III	Negative	Negative
Forward regeneration	IV	Negative	Positive

	Quadrant	DC current	AC sequence
Forward motoring	I	Positive	Positive
Backward regeneration	II	Negative	Negative
Backward motoring	III	Positive	Negative
Forward regeneration	IV	Negative	Positive



Induction Machine



Induction Machine

Overview

- Commutatorless motor drives offer a number of advantages over conventional DC motor drives for electric propulsion, as low cost, high power factor, reliable operation.
- An AC induction motor is one of the simplest and most reliable types of electric AC motor, with the power ranging from a few watts to many kilowatts.
- There are many different ways to design and control such motors, from low cost with a single-phase input to high accuracy with a polyphase approach, and this type of system is the most widely used motor across the consumer and industrial markets.

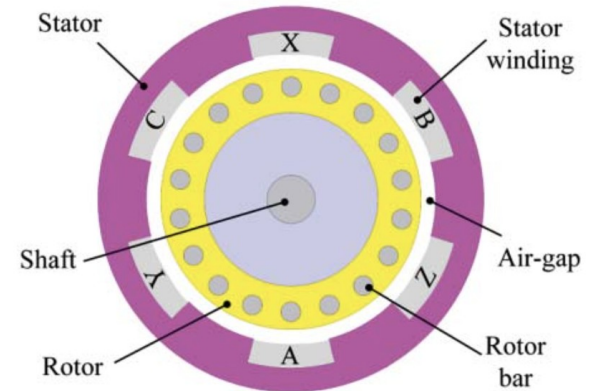
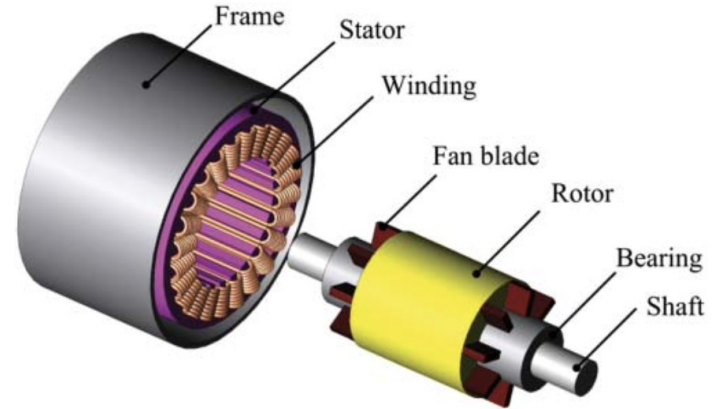


Induction Machine Structure

The most common type of IMs is the **squirrel-cage IM**.

It consists of:

- a **stator** incorporated with three-phase armature windings;
- a **rotor** incorporated with cage bars that are short-circuited by two end-rings;
- two end-**bearings** to support the rotor;
- a machine **frame** with two end-bells to house the machine.





Induction Machine Working principle

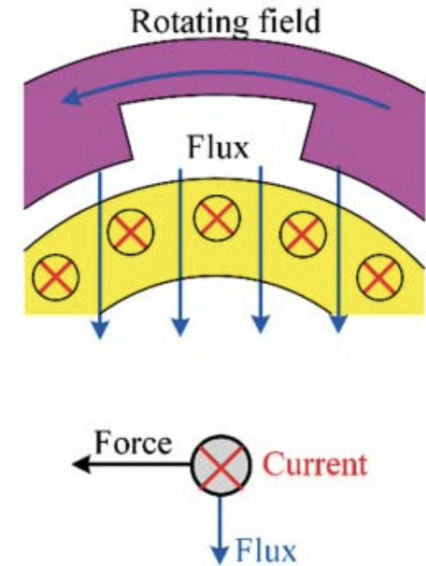
The stator of a three-phase induction machine contains a series of windings to which three-phase electrical power is applied.

These windings establish **a rotating magnetic field** in the air gap between the stator and the rotor.

The stator field can be seen as a series of north-south poles revolving around the circumference of the stator.

The rotation of the stator magnetic field occurs at a fixed speed called **synchronous speed** and linked to the supply frequency f .

In IMs, **the angular speed of the rotor is lower than the synchronous speed of the magnetic field** generated by the stator windings, hence called asynchronous.



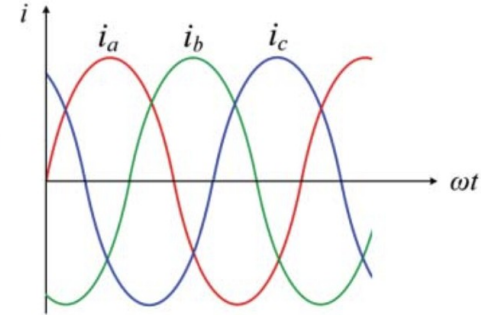
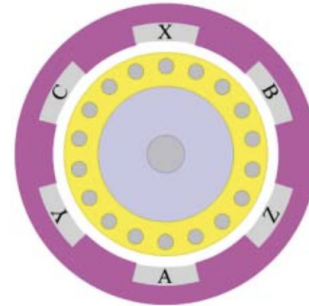


Induction Machine Working principle

The rotor windings are immersed in this rotating magnetic field. Since the rotor turns slower than the stator field, the magnetic flux linkage in the rotor windings varies.

The rotating magnetic field **induces currents in the rotor** by magnetic induction (Faraday's law).

These induced currents, in turn, **generate a rotor magnetic field**, which opposes flux variations; it interacts with the stator field, **generating a torque** on the rotor windings which causes the rotor to rotate.





Induction Machine Working principle

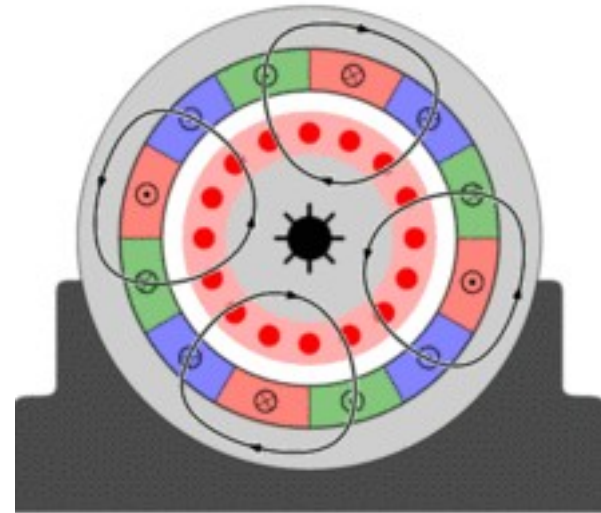
The link between synchronous speed, supply frequency f and the number p of poles (or p_c of pole pairs per phase) is expressed by the relation:

$$n_s = \frac{120f}{p} = \frac{60f}{p_c}$$

Since the relative motion between the rotating field and the rotor is the key for torque production, the **slip** s is defined as:

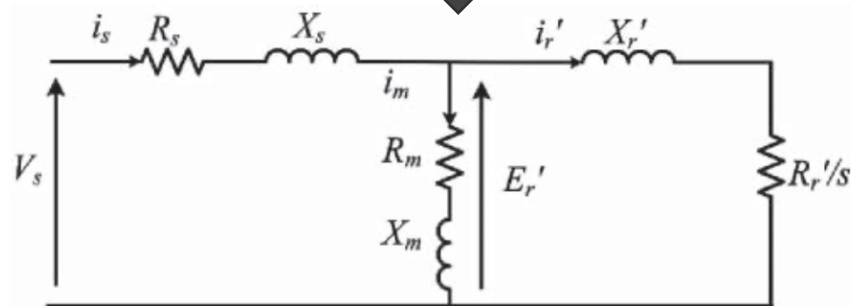
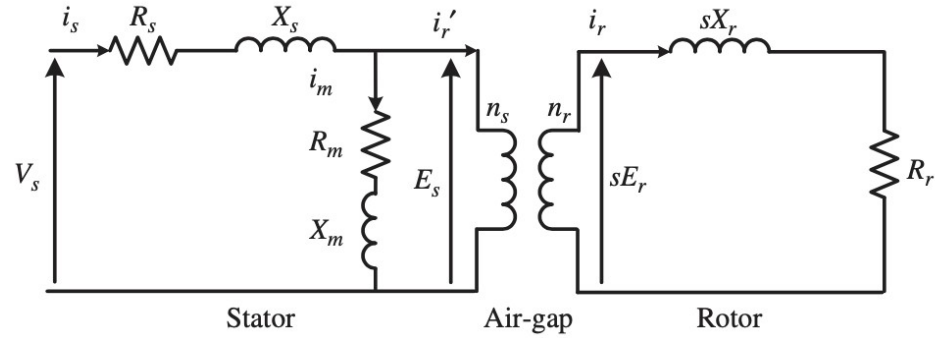
$$s = \frac{\omega_s - \omega_r}{\omega_s}$$

Where ω_s represents the synchronous speed and ω_r the rotor speed.





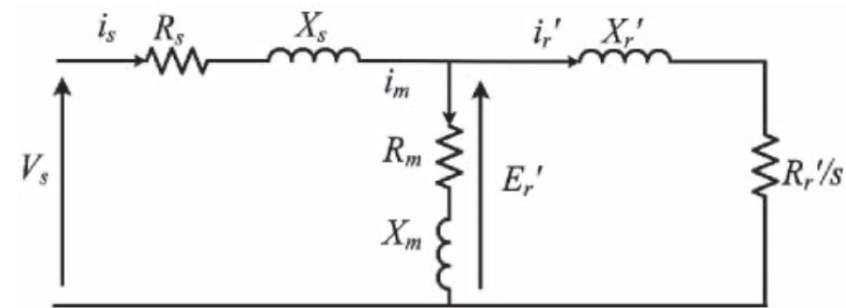
Induction Machine Modeling





Induction Machine Modeling

- R_s and X_s are respectively the winding resistance and leakage reactance in the stator;
- R_r and X_r are respectively the winding resistance and leakage reactance in the rotor at standstill;
- E_s and E_r are respectively the induced EMF in the stator and induced EMF in the rotor at standstill;
- R_m and X_m are respectively the core-loss resistance and magnetizing reactance in the stator;
- k is the ratio of the number of turns of stator winding to the number of turns of rotor winding.





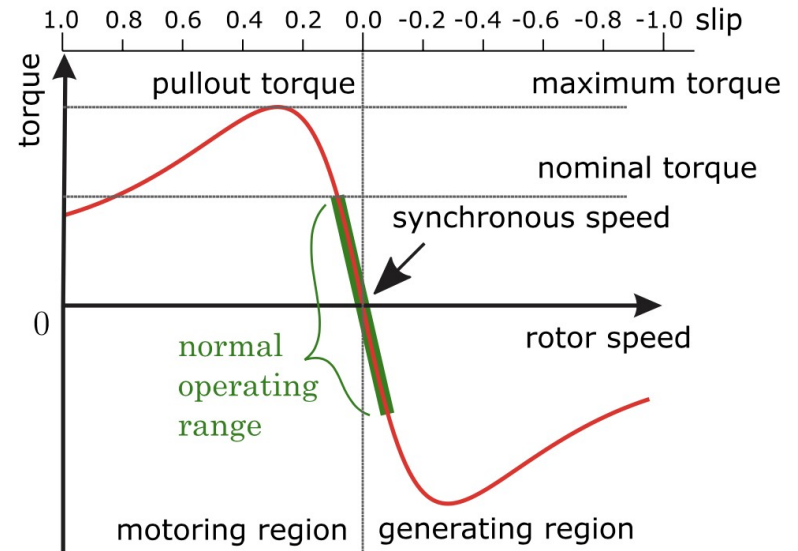
Induction Machine Torque

At **synchronous speed**, no torque is produced by the motor, because no current is induced in the rotor windings. If the rotor rotates at a lower speed than the stator field, the motor runs in **motoring mode**.

Otherwise, the motor operates in **generator mode**. The maximum torque, which can be obtained at the motor shaft, is called **pullout torque**.

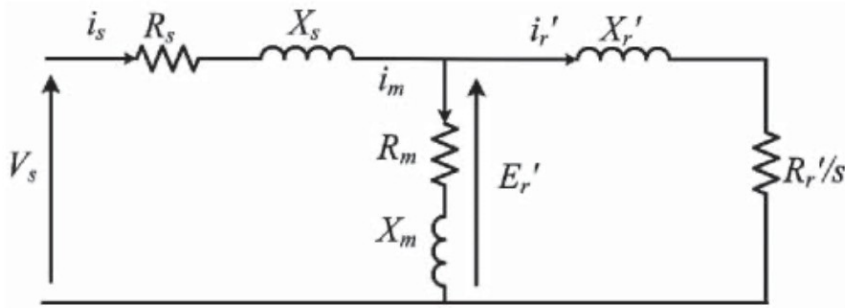
The nominal torque is conventionally half of the pullout torque.

The motor usually operates in the **linear region** between the positive and the negative nominal torque.





Induction Machine Modeling



$$Z_s = R_s + jX_s \quad I_s = \frac{V_s}{(Z_s + Z_m || Z_r')}$$

$$Z_m = R_m + jX_m$$

$$Z_r' = \frac{R_r'}{s} + jX_r'$$

$$I_r' = \frac{Z_m}{Z_m + Z_r'} I_s$$

$$P_s = \frac{3R_r'}{s} I_r'^2$$

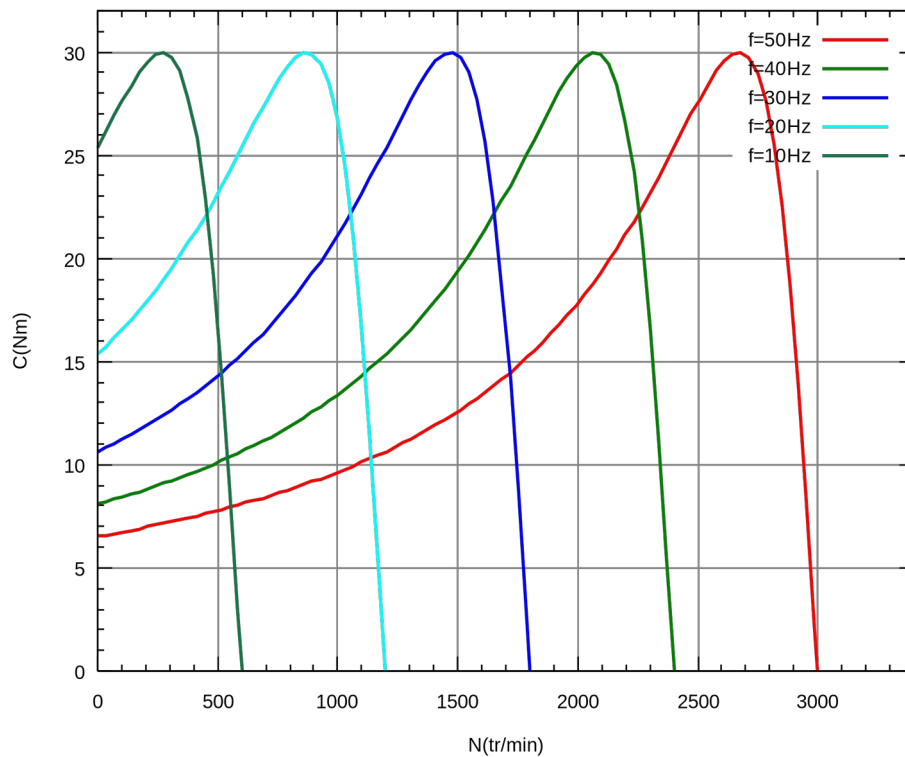
$$\Delta P_{Cu_r} = 3R_r' I_r'^2$$

$$P_m = P_s - \Delta P_{Cu_r} = 3 \frac{1-s}{s} R_r' I_r'^2$$

$$C = \frac{P}{\omega_r} P_m = \frac{3P}{\omega_r} R_2' \frac{1-s}{s} I_r' = \frac{3P}{\omega} \frac{R_2'}{s} I_r'^2$$



Induction Machine Speed Control





Induction Machine **Squirrel cage**

Because of high cost, need for maintenance, and lack of sturdiness, the wound-rotor induction motor is less attractive than the squirrel-cage counterpart, especially for electric propulsion in electric vehicles (EVs). Hence, the **squirrel-cage** induction motor is loosely named as the induction motor for EV propulsion.

In addition to the common advantages of commutatorless motor drives, the induction motor drives possess the definite advantages of **low cost** and **ruggedness**. These advantages can generally outweigh their major disadvantage of **control complexity** and facilitate this technology being widely accepted for EVs.





Permanent Magnet Brushless Machine



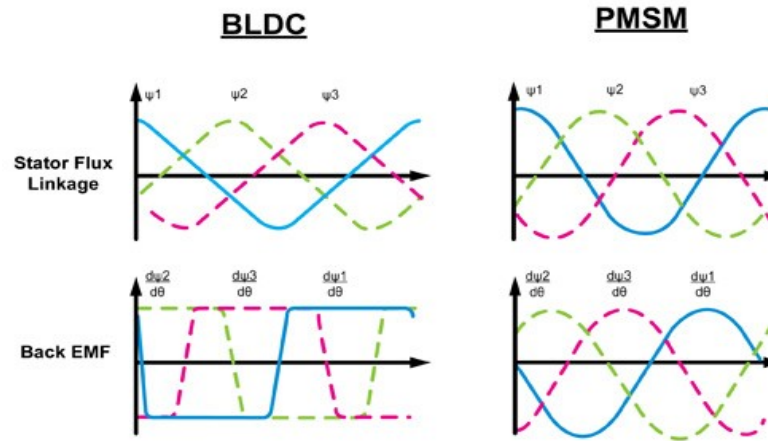
- Among various types of motor drives, permanent magnet (PM) brushless motor drives, especially the PM synchronous motor drive, are currently the most attractive motor drives for electric vehicle (EV) propulsion.
- Their key features, namely **high power density** and **high efficiency**, are attributed to the use of high-energy PM material. Actually, they are becoming dominant in the market share of EV motor drives. However, these PM brushless motor drives still have some shortcomings such as **high cost** and **thermal instability** of the PM material.
- The family of PM brushless machines has two main members: the **PM synchronous machine** and the **PM BLDC machine**. The PM synchronous machine has been widely used for EV propulsion; meanwhile, the PM BLDC machine is becoming attractive.



PM Brushless Classification

Brushless synchronous motors can be categorized as follows according to **the shape of the back emf** developed at the air gap:

- **Trapezoidal**, the machine is powered by CSI inverter, BLDC (Brushless DC) machine;
- **Sinusoidal**, the machine is powered by VSI inverters with sinusoidal voltages (PMSM).



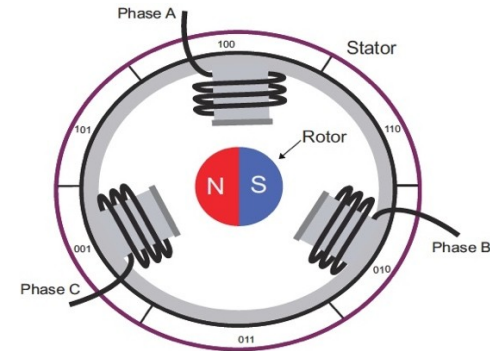
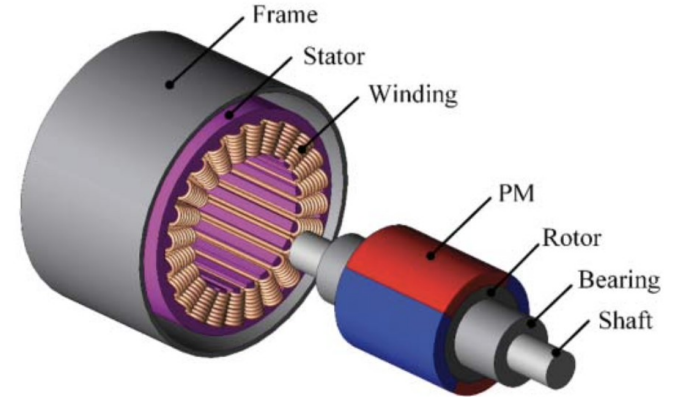


PM Brushless Structure

The basic structure of the PM brushless machine, including both the PMSM and BLDC types, consists of:

- a stator housing a **winding**, typically three-phase connected in a star, through which the motor itself is powered;
- a rotor equipped with **permanent magnets** capable of establishing a magnetic field in the air gap quite similar to that produced by the excitation winding placed on the rotor of normal synchronous motors.

Compared with the induction machine, it is relatively simpler because of the **absence of cage bars** and end-rings.





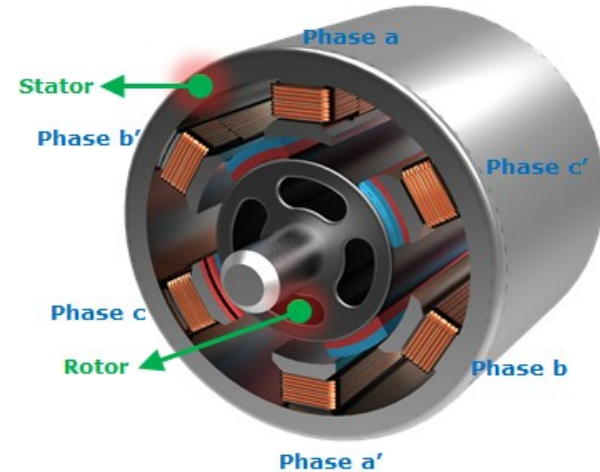
PMSM

Working principle

In the machine, the rotor rotates **at the same speed** as the rotating magnetic field of the stator, so the mechanical speed of the rotor coincides with the synchronous speed.

$$n_s = \frac{60f}{p}$$

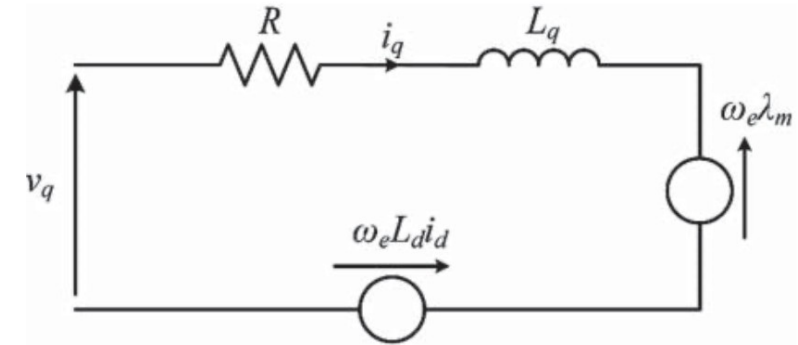
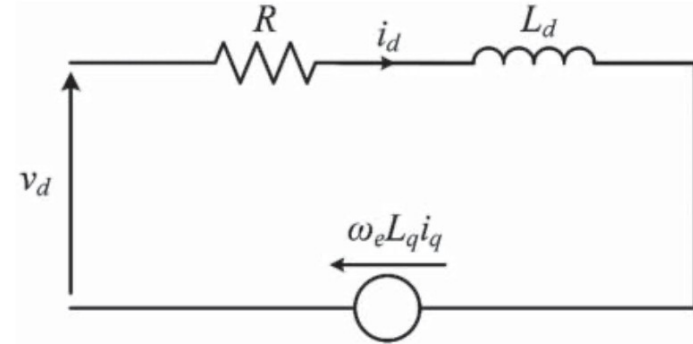
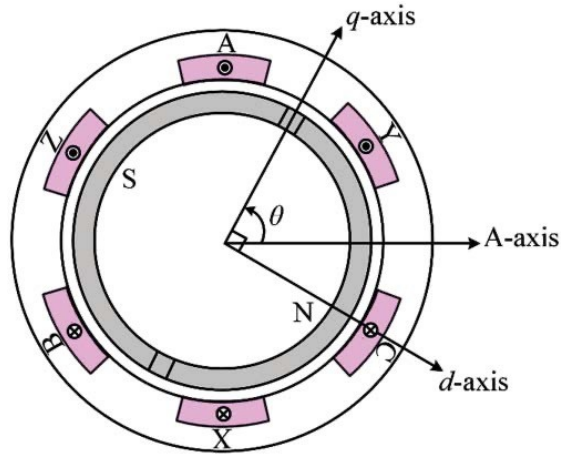
The principle of PM synchronous machine is based on the interaction of **sinusoidal back electromotive force (EMF)** waveforms and **sinusoidal armature (stator) current** waveforms.





PMSM Modeling

The machine model is expressed about the rotor d-q reference frame.





PMSM

Modeling

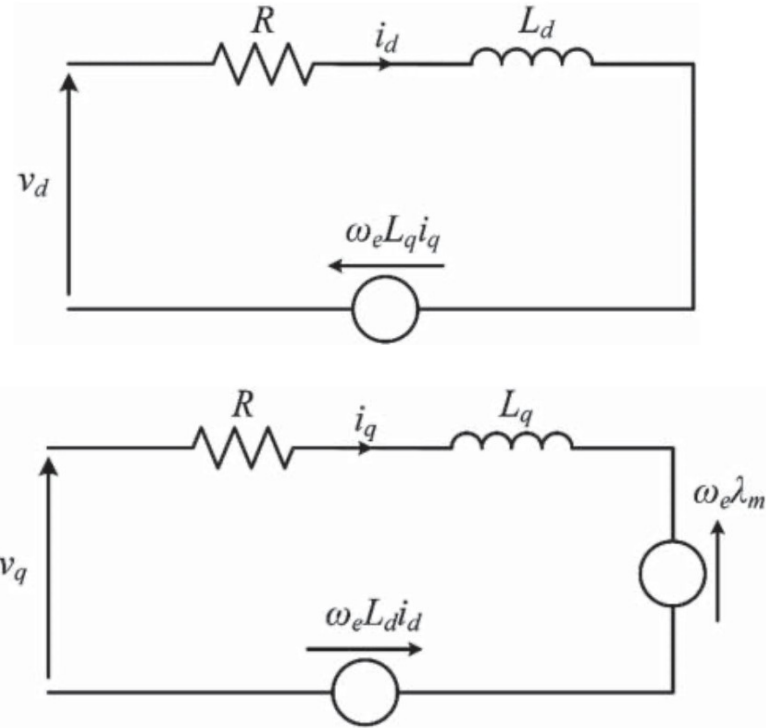
$$v_d = Ri_d + p\lambda_d - \omega_e \lambda_q$$

$$v_q = Ri_q + p\lambda_q + \omega_e \lambda_d$$

Where v_d and v_q are, respectively, the d -axis and q -axis components of stator voltages, i_d and i_q are, respectively, the d -axis and q -axis components of stator currents, and λ_d and λ_q are the d -axis and q -axis components of flux linkages, which can be expressed as:

$$\lambda_d = L_d i_d + \lambda_m$$

$$\lambda_q = L_q i_q$$

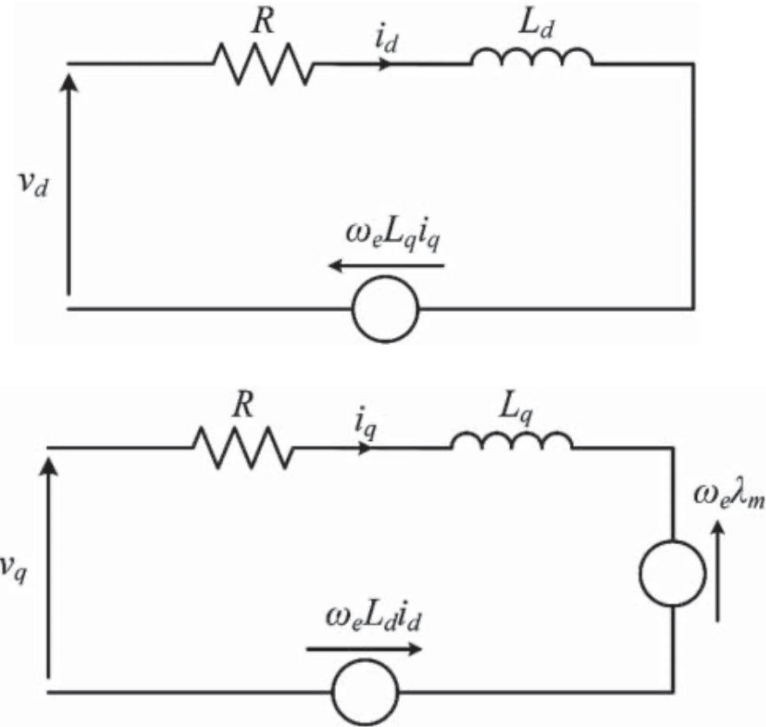




The electromagnetic developed output torque is equal to:

$$T_e = \frac{3}{2}P(\lambda_m i_q + (L_d - L_q)i_d i_q)$$

The developed torque is composed of two torque components. The first component corresponds to the **reaction torque** between the PM flux and q -axis armature current, while the second component corresponds to the **reluctance torque** due to the difference between d -axis and q -axis inductances.



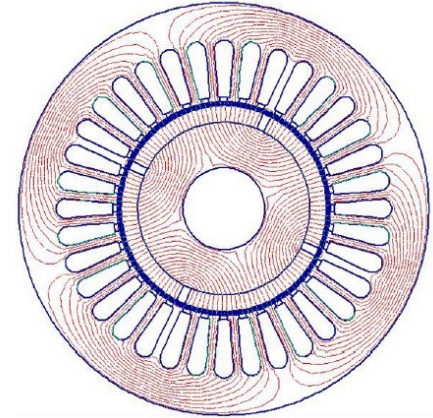


BLDC

Working principle

The principle of a BLDC machine is based on the interaction of **trapezoidal back EMF waveforms** and **rectangular armature current waveforms**.

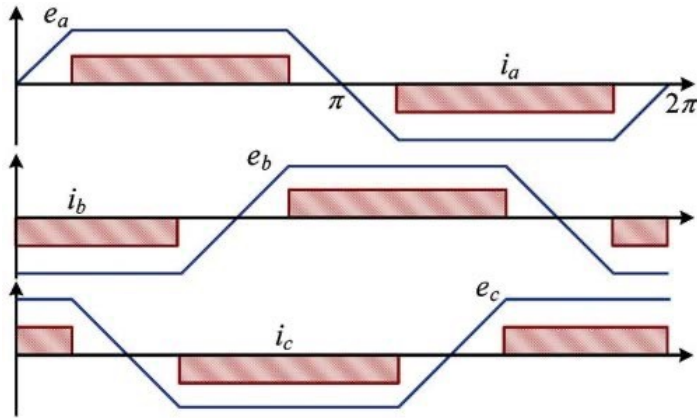
BLDC motors are also referred to as **electronically commutated motor**. As with mechanical commutation, the electronic commutation helps in achieving unidirectional torque similar to a conventional DC machine.



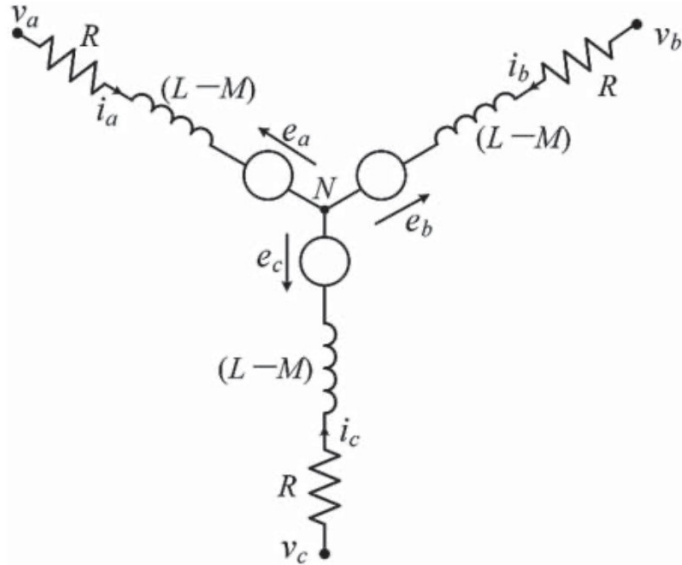


BLDC

Modeling

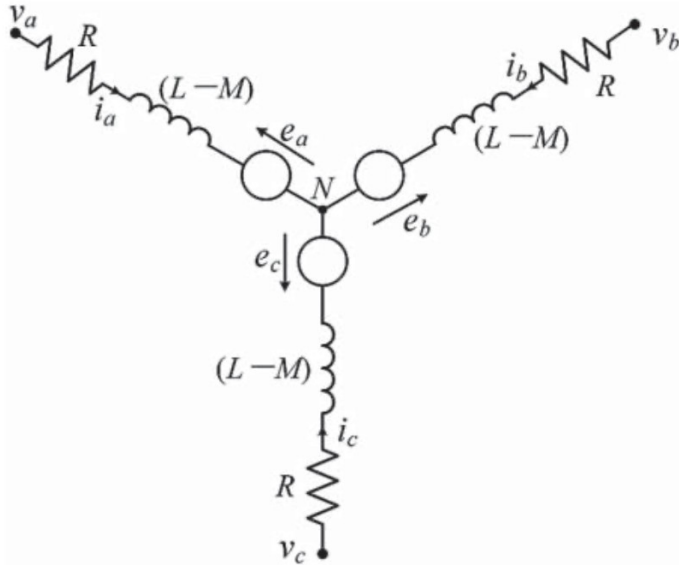


In a BLDC machine the back EMF waveforms are trapezoidal, rather than sinusoidal. Thus, **the $d-q$ modeling** technique developed for the PM synchronous machine **is not applicable**. Since the flux linkage is nonsinusoidal, it is necessary to derive the model of the PM brushless machine in terms of its **state variables**.



$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + p \begin{bmatrix} L_{aa} & L_{ab} & L_{ac} \\ L_{ba} & L_{bb} & L_{bc} \\ L_{ca} & L_{cb} & L_{cc} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix}$$

- v_a , v_b , and v_c are the instantaneous three-phase stator voltages;
- i_a , i_b , and i_c are the instantaneous three-phase stator currents;
- e_a , e_b , and e_c are the instantaneous three-phase back EMFs;
- R is the armature resistance;
- L_{ij} ($i = a, b, c$ and $j = a, b, c$) are the mutual inductances which are symmetrical.



The developed torque T_e is the output power divided by the mechanical speed, which is given by:

$$T_e = \frac{P_e}{\omega_r} = P \frac{(e_a i_a + e_b i_b + e_c i_c)}{\omega_e}$$

As each phase back EMF is a **bipolar trapezoidal waveform** displaced by 120° (electrical), each current waveform needs to be 120° on positive and 120° on negative within a period to ensure the synchronism and hence produce the desired steady torque without torque pulsations.



SRM

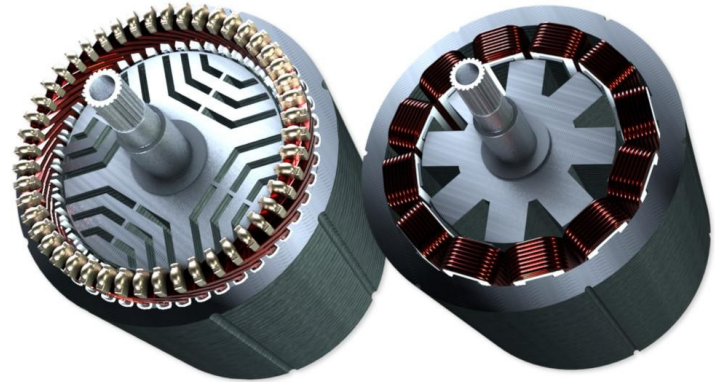
Working principle

The **Switched Reluctance Motor (SRM)** is an electric motor that runs by reluctance torque.

The rotor has no magnets or coils attached. It is a solid salient-pole rotor made of **soft magnetic material**.

When power is applied to the stator windings, the rotor's magnetic reluctance creates a force that attempts to align the rotor pole with the nearest stator pole.

In order to maintain rotation, an electronic control system switches on the windings of successive stator poles in sequence so that magnetic field of the stator leads the rotor pole, pulling it forward.

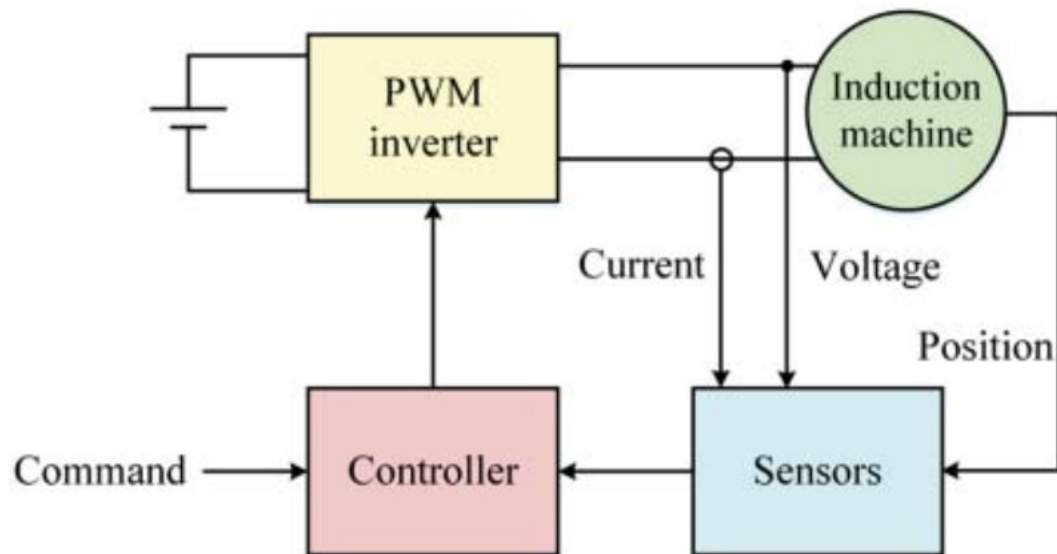




Control Strategies



Control strategies **Block Diagram**



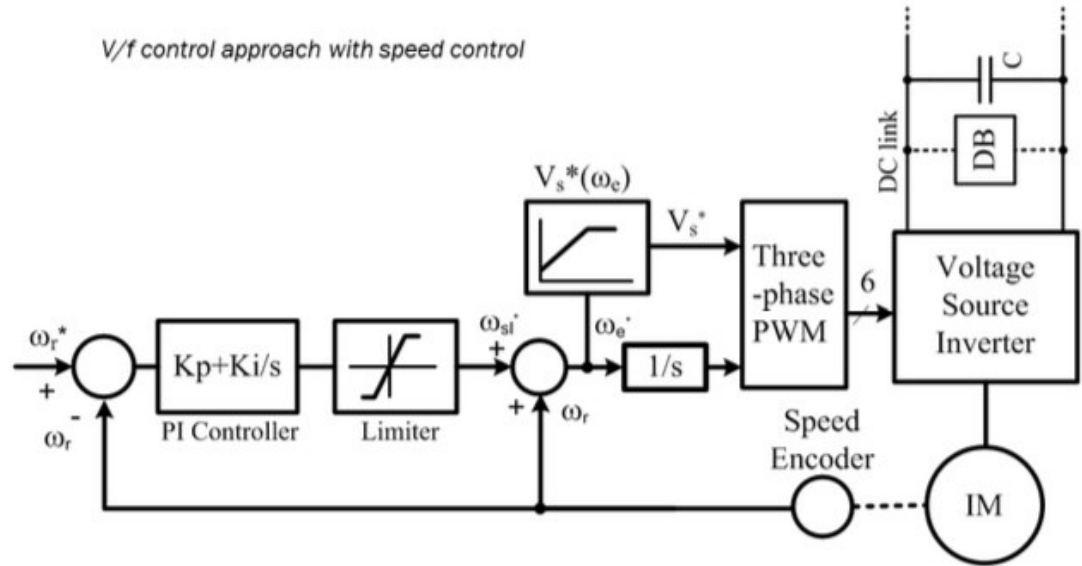


Control strategies **V/f**

In the past, mainly the rather simple **voltage-to-frequency (V/f) control** was used and it can still be found in industrial drives.

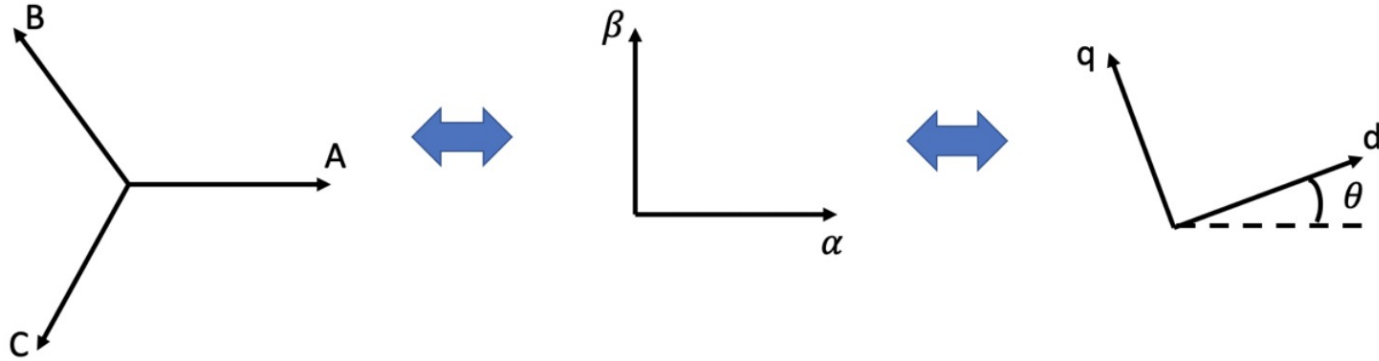
This method is a **scalar method**, that is, it operates just on the module (and not the phase) of the control quantities involved.

The main drawback is its **limited dynamic performance**.





Control strategies Park-Clarke transformations

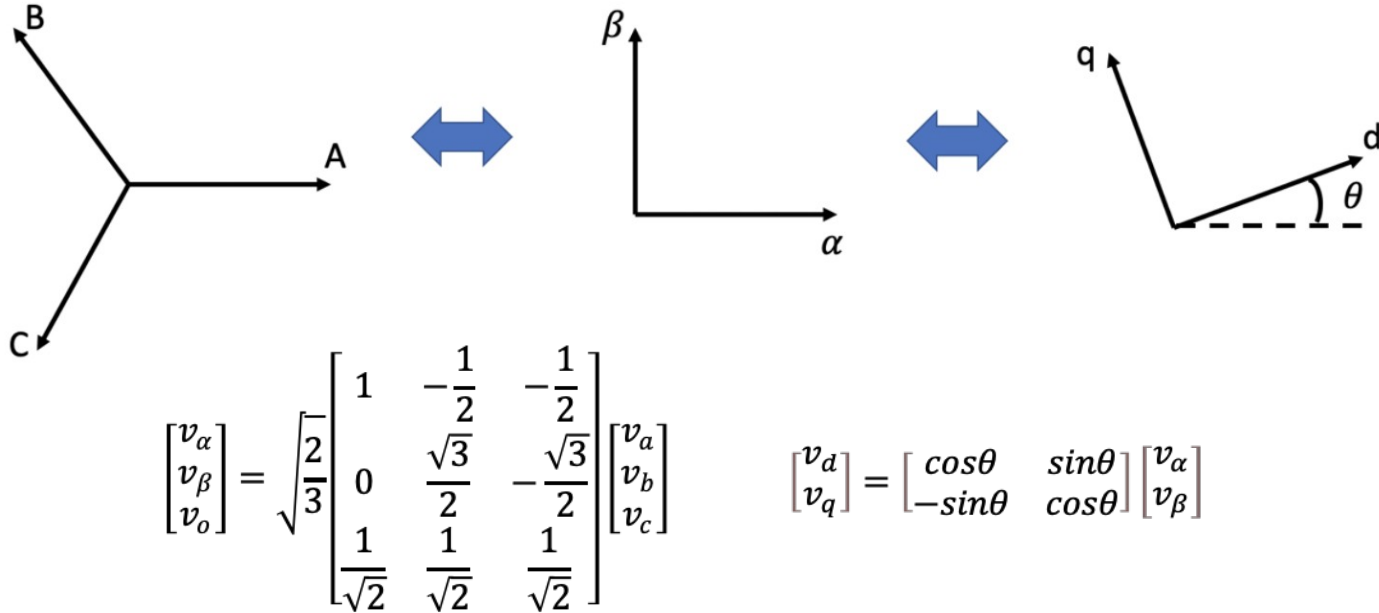


With the letters **ABC** is denoted the three-phase stator power supply system, with $\alpha\beta$ the equivalent two-phase stator (and therefore stationary) reference system and with **dq** the reference system joined with the rotor and therefore rotating at rotor speed. The transformation from **ABC** to $\alpha\beta$ is called **Park transformation**, while $\alpha\beta$ to **dq** is called **Clarke transformation**.



Control strategies

Park-Clarke transformations





Control strategies

FOC

Another control strategy for inverter-fed variable-speed drives is the “**Field Oriented Control**” (FOC).

The time-dependent instantaneous three-phase system values, such as currents and voltages, are transformed into an orthogonal two-dimensional coordinate system by means of the Park transformation. This coordinate system rotates with the same frequency as the three-phase values.

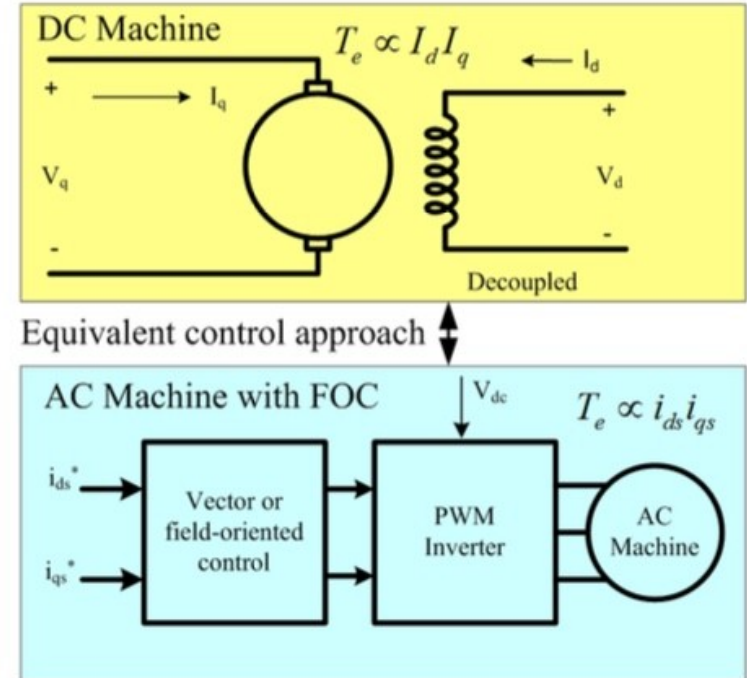
From this, **the three AC quantities are reduced to two DC quantities** which, in general, are much simpler to control. The two DC quantities can be represented as a “space vector” in the two-dimensional coordinate system. This transformation results in a controllability of torque of the induction machine that is similar to the DC machine.



Control strategies **FOC**

This technique controls the variable of interest **in module and phase** with the aim of separating, as occurs in the direct current machine, the current components that contribute to the flux and torque.

With this type of regulation it is possible to obtain **rapid response and good adaptation** by the machine, whose operation, moreover, can be extended to the **four quadrants**.





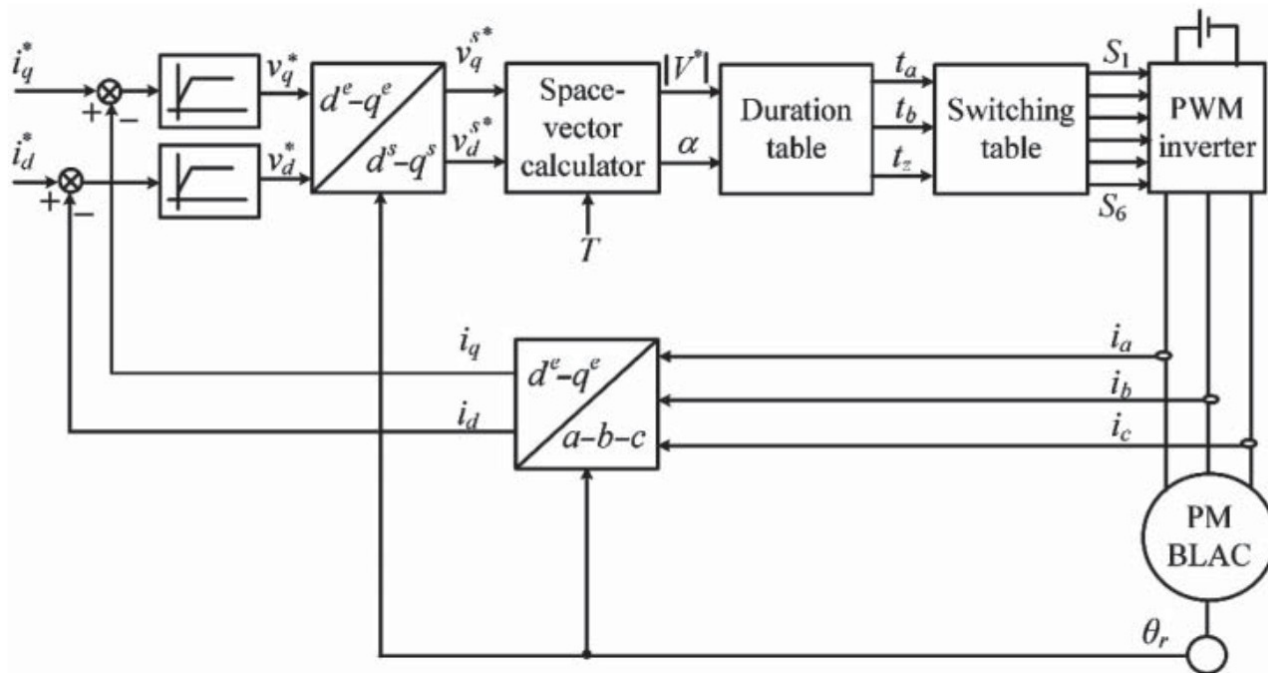
Control strategies **FOC**

The advantages of field-oriented control can be summarized as follows:

- **High efficiency** obtained by aligning the stator and rotor flows to generate excellent torque production by the machine;
- **High dynamic response** obtained by controlling the quantities (voltages, currents and fluxes) in module and phase;
- **Low torque ripple**, as the stator field is continuously updated according to the position of the rotor field;
- **Independent control of flux and torque**, as in direct current machines;
- **High dynamic range** through weakening operation;
- **High torque and low starting current.**



Control strategies Space Vector Modulation





Conclusions



Among the three main types of commercially available EV motor drives, namely the DC, induction, and PM brushless, the PM brushless motor drives take the definite advantages of **higher efficiency, higher power density, and higher torque density**, which are highly desirable for EV propulsion. Currently, there is no doubt that the PM brushless motor drives are **the preferred technology** for EVs.

Between the two members of PM brushless motor drives, the PM synchronous motor drive is **more mature** than the PM BLDC motor drive while they have their individual merits. That is, the PM synchronous motor drive has relatively **higher efficiency** and **lower torque ripple**, whereas the PM BLDC motor drive has relatively **higher power density** and **higher torque density**.

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